

Bo_Bichette_BIP_2023

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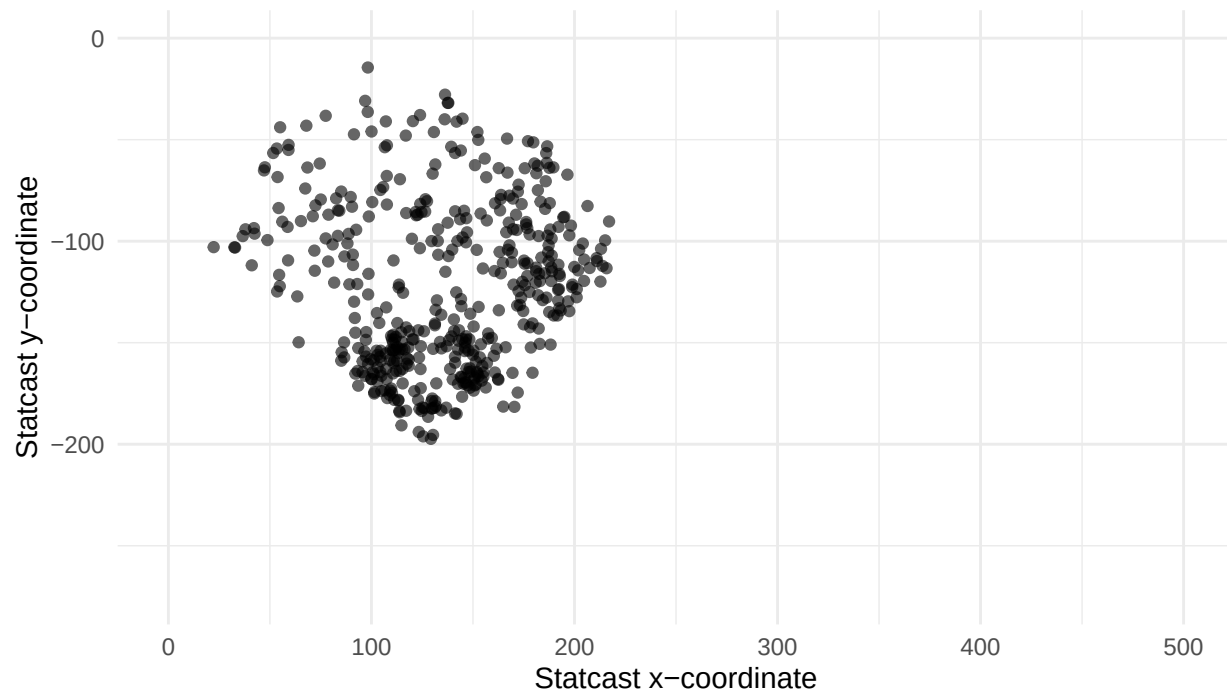
2026-04-21

```
## Response [https://baseballsavant.mlb.com/statcast_search/csv?all=true&hfPT=&hfAB=&hfBBT=&hfPR=&hfZ=&
##   Date: 2026-04-22 02:31
##   Status: 200
##   Content-Type: application/download; charset=utf-8
##   Size: 1.48 MB
## <ON DISK>  C:\Users\4reve\AppData\Local\Temp\RtmpScaKDD\filef3814962743.csv

## # A tibble: 6 x 11
##   player_name game_date events stand batter pitcher hc_x hc_y launch_speed
##   <chr>       <date>    <chr>  <chr>  <dbl>  <dbl> <dbl> <dbl>      <dbl>
## 1 Bichette, Bo 2023-10-01 field_o~ R      666182 682052 124.   81.6      86.2
## 2 Bichette, Bo 2023-10-01 field_o~ R      666182 682052 156.   172.     100.
## 3 Bichette, Bo 2023-10-01 triple  R      666182 671737 213.   104.     97.1
## 4 Bichette, Bo 2023-09-30 field_o~ R      666182 681867 141.   185.     58.3
## 5 Bichette, Bo 2023-09-30 double  R      666182 606965 67.4   74.1     106
## 6 Bichette, Bo 2023-09-30 field_o~ R      666182 542888 177.   93.6     90.3
## # i 2 more variables: launch_angle <dbl>, hit_distance_sc <dbl>

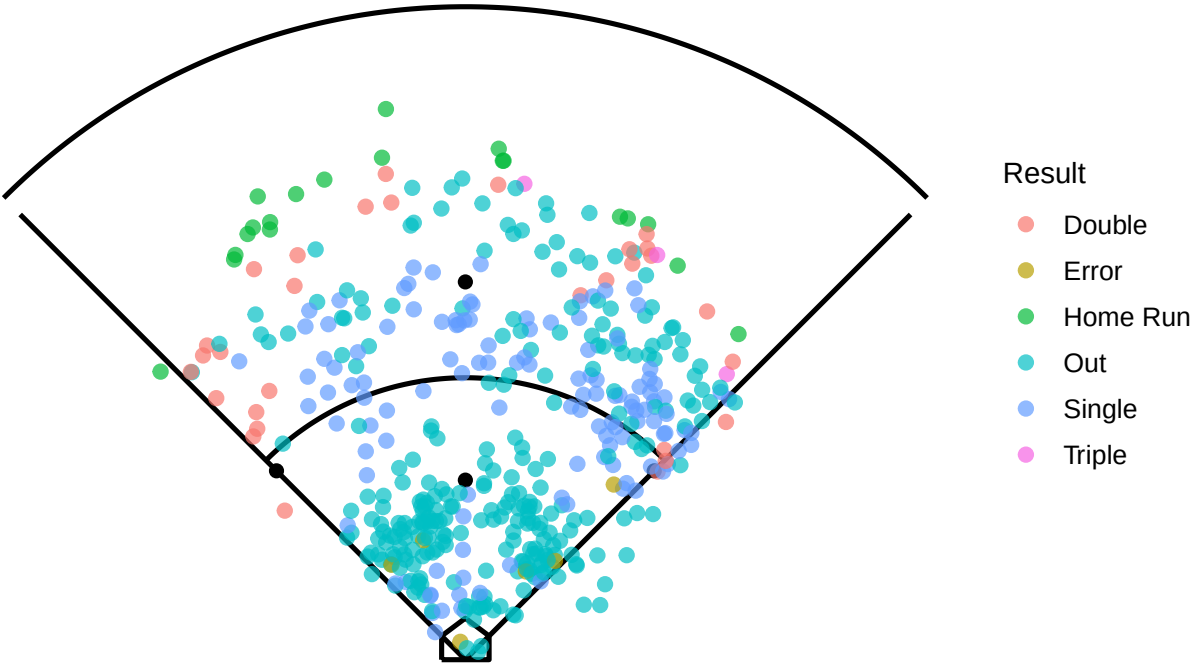
## [1] 457
```

Bo Bichette 2023 Spray Chart



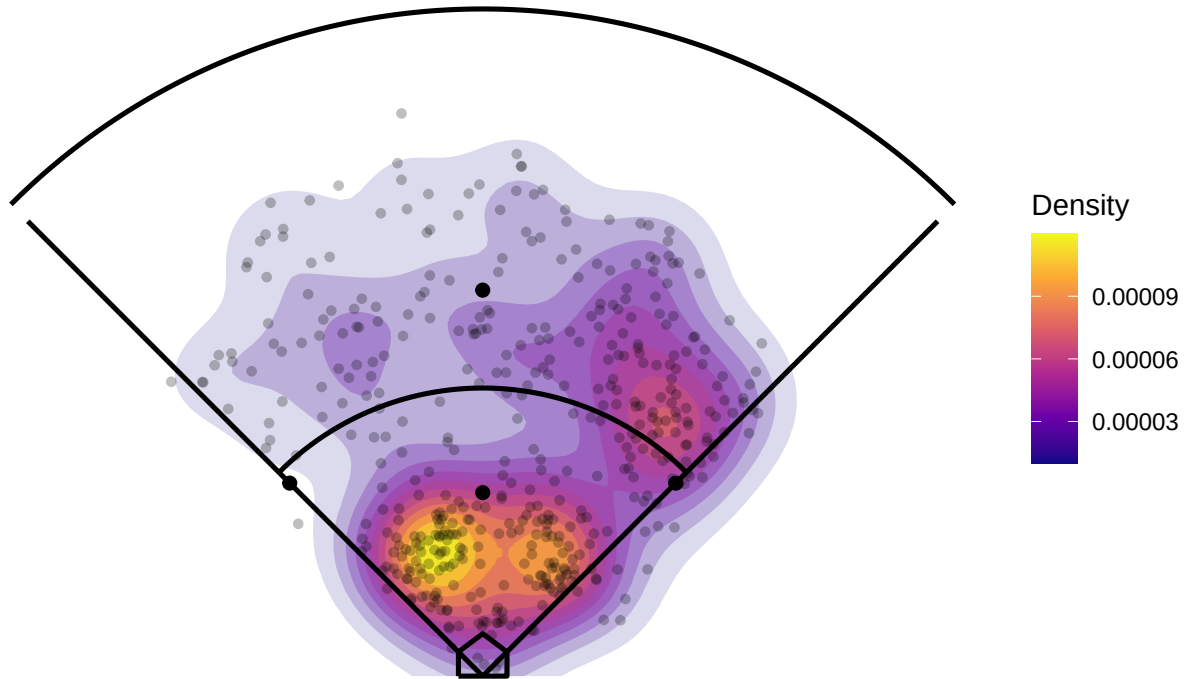
Bo Bichette 2023 Spray Chart

Balls in play classified by result



Bo Bichette 2023 Kernel Density Map

Density of balls in play



```
## # A tibble: 9 x 5
##   y_zone x_zone zone_9      n proportion
##   <chr>  <chr>  <chr>    <int>     <dbl>
## 1 Deep   Left    Deep_Left    55     0.120
## 2 Deep   Center  Deep_Center   40     0.0875
## 3 Deep   Right   Deep_Right    57     0.125
## 4 Medium Left    Medium_Left   39     0.0853
## 5 Medium Center  Medium_Center 37     0.0810
## 6 Medium Right   Medium_Right   76     0.166
## 7 Shallow Left    Shallow_Left   59     0.129
## 8 Shallow Center  Shallow_Center 75     0.164
## 9 Shallow Right   Shallow_Right  19     0.0416
```

```
##
## Chi-squared test for given probabilities
##
## data: zone_counts_9$n
## X-squared = 55.173, df = 8, p-value = 4.086e-09
```

```
##           zone_9 observed expected
## 1      Deep_Left      55    50.78
## 2      Deep_Center      40    50.78
## 3      Deep_Right      57    50.78
## 4      Medium_Left      39    50.78
## 5      Medium_Center      37    50.78
```

## 6	Medium_Right	76	50.78
## 7	Shallow_Left	59	50.78
## 8	Shallow_Center	75	50.78
## 9	Shallow_Right	19	50.78

The p-value is very low $4 * 10^{-9}$ we reject the null hypothesis that balls are evenly distributed across the 9 zones identifying a non-uniform distribution.

Bo Bichette 2023 Spray Chart with 9-Zone Grid

in play divided into shallow, medium, and deep thirds across left, center, and right field

